MIL-T-52820A(ME)
31 October 1979
SUPERSEDING
MIL-T-52820(ME)
8 July 1974

MILITARY SPECIFICATION

TABLE, DRAWING, STEEL

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a steel drawing table.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-S-92 - Screw, Machine: Slotted, Cross-Recessed or Hexagon Head.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| | |
|---|---|
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-F-320 | - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes. |
| PPP-T-60 | - Tape: Packaging, Waterproof. |

MILITARY

| | |
|---|---|
| MIL-W-530 | - Webbing, Textile, Cotton, General Purpose, Natural or in Colors. |
| MIL-B-543 | - Buckles, Tongueless and Web Strap. |
| MIL-T-704 | - Treatment and Painting of Materiel. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED. STD. No. 595 | - Colors. |

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| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |
| MS35425 | - Nut, Plain, Wing, UNC-2B. |
| MS35826 | - Hinge, Butt, Continuous (Piano) .060 Thickness .187 Pin Dia. |
| MS39123 | - Drawing Boards, Type I. |
| MS51939 | - Loop, Strap Fastener (Footmans Loop). |



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with Method 6071 of FED-STD-228 duster shall be measured in accordance

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| A 36 | Structural Steel. |
| A 366 | Steel, Carbon, Cold-Rolled Sheet, Commercial Quality. |
| A 519 | Seamless Carbon and Alloy Steel Mechanical Tubing. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The table shall be as shown in Figures 1 and 2 and as specified herein.

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.2.1 Sheet steel Sheet steel shall conform to ASTM A 366.

3.2.2 Tube steel. Tube steel shall conform to ASTM A 519.

3.2.3 Angle steel. Angle steel shall conform to ASTM A 36.



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3.3 Fabrication.

3.3.1 Storage compartment. The storage compartment shall be cold rolled commercial quality sheet steel and shall be provided with 2 lengthwise partitions, equally spaced; 4 widthwise partitions, equally spaced; drawer; and an ink tray.

3.3.1.1 Drawer. The drawer shall ride within channels welded to the compartment body and shall be fitted with a latch and drawer pull.

3.3.1.1.1 Drawer latch. Each drawer shall have an automatic latch which is released by a finger-controlled mechanism. The finger control may be a button or lever and shall be located so that it can be released when the user's hand is in position on the drawer pull. The latch shall automatically hold the drawer in a closed position. The latch shall prevent the drawer from opening, even when the loaded drawer is subjected to the shock and vibration encountered when the table is mounted in a motorized van traversing rough terrain, unless the latch is released by the finger control. The latching mechanism, except for the locking hook, shall be concealed between the drawer pans. The catch may be formed from the drawer suspension member of the front vertical reinforcing member in such a way as to positively and automatically engage the hook, and in a manner to prevent shearing or tearing of the catch by the hook. The button or lever shall be fabricated from corrosion-resistant metal to assure free action.

3.3.1.2 Ink tray. The ink tray shall be welded to the inside bottom of the drawer.

3.3.2 Drawing board and strip. The drawing board shall conform to MS39123-5. The strip shall be 0.75 by 0.75 by 42 inches, and shall be the same material and finish as the drawing board.

3.3.3 Hinge. The hinge shall conform to MS35826-10D, 42 inches long.

3.3.4 Adjustable legs. The legs shall be tube steel conforming to ASTM A 519, seamless, normalized, and square shape. Each leg shall consist of an inner tube and an outer tube. The dimensions of the inner tube shall be as follows: Length, 26 inches; outside diameter, 2 inches; and wall thickness, 0.036 inch. The dimensions of the outer tube shall be as follows: Length, 8 inches; outside diameter, 2.04 inches; and wall thickness, 0.036 inch. Each outer tube shall be welded



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to the bottom of the storage compartment and shall have a 0.50-inch bolthole, drilled with center 1 inch from the end. Each inner tube shall have three 0.50-inch boltholes, drilled with centers 1, 4, and 7 inches from the end, to accommodate height adjustment bolts. The left rear inner tube shall have four 0.203-inch boltholes, two drilled with centers 2 and 4.875 inches above the top of and on the same side as the leg brace, and two drilled with centers 2 and 4.875 inches above the top of and on the inward side adjacent to the leg brace. These four boltholes are for attachment of the footmans loops.

3.3.5 Side braces. Each side brace shall be 2- by 2- by 0.188-inch angle comforming to ASTM A 36. Upper side braces shall be welded to the storage compartment and to the outer leg tubes. Lower side braces shall be welded to the inner leg tubes.

3.3.6 Swivel stool and bracket. The stool bracket shall be tube steel conforming to ASTM A 519, seamless, normalized, round shape, with a 2-inch outside diameter. The seat shall be wood or steel and shall be provided with a rotary mechanism. The mechanism shall provide free rotation on a horizontal plane with height adjustable by hand. The seat mounting shall not distort under normal load. The spindle shall permit a height adjustment of not less than 3 inches. The hub of the rotary mechanism shall be welded to the bracket. The bracket shall be attached to the desk by hinges, which shall be welded or bolted to the inner tube of the left front leg.

3.3.7 Footmans loops. Footmans loops shall conform to MS51939-1, Type V, and shall be bolted to the inner tube of the rear left leg.

3.3.8 Straps. Straps shall conform to MIL-W-530, Type II, Class 1, 1.25 x 11-inch size and shall be fastened to the left rear leg by footmans loops.

3.3.9 Buckles. Buckles shall conform to MIL-B-543, Type II, Style 4, Class 1, size 1.50 inch.

3.3.10 Bolts. Bolts used in leg adjustment shall conform to FF-S-92, Type I, Style 1s, 0.500 - 13 UNC-2A, length 2.50 inches.

3.3.11 Wing nuts. Wing nuts shall conform to MS35425-44.

3.4 Marking. The drawing table shall be identified in accordance with MIL-STD-130.



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3.5 Treatment and painting. All metal portions of the table, except the steel end cleats, drawing board hinges, threaded part of the swivel stool, and the metal tip on the strap, shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Color shall be Green conforming to FED. STD. No. 595, Color No. 24525.

3.6 Workmanship.

3.6.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of materials shall be done by methods that will not cause damage to the material. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Surfaces shall be free from burrs, sharp edges, and sharp corners.

3.6.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strengtn of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

3.6.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 0.4 percent defective for major defects and 4.0 percent defective for minor defects.

4.4 Inspection procedure.

4.4.1 Examination. The table shall be examined as specified herein for the following defects:

Major

101. Table not as shown in Figure 1.
102. Material not as specified.
103. Construction not as specified.
104. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or not legible.
202. Treatment and painting not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.



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4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

| | |
|---|---|
| 105. | Materials or containers not as specified. Each incorrect material or container shall be considered one defect. |
| 106. | Stool threads not lubricated as specified for Level A. |
| 107. | Table not lowered on legs for Level A. |
| 108. | Stool not secured under table for Level A. |
| 109. | Table drawer not secured in closed position for Level A. |
| 110. | Table not wrapped and wrap not secured with tape as specified for Level A. |
| 111. | Closure and strapping not as specified for Level A or Level B. |
| 112. | Blocking and bracing not as specified for Level A or Level B. |
| 113. | Marking illegible, incomplete, incorrect, or missing. |

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The threaded part of the swivel stool shall be lubricated with the manufacturer's recommended lubricant. The table shall be lowered on the legs and the stool secured under the table. The table drawer shall be secured in the closed position. The table shall be wrapped with a polyethylene material, not less than 1.50 mil in thickness. The wrap shall be secured with tape conforming to PPP-T-60, Type III or IV.

5.1.2 Commercial. The tables shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).



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5.2.1 Level A. The tables shall be packed in boxes conforming to PPP-B-601, Overseas Type, Grade B, or PPP-B-621, Class 2. The tables shall be blocked and braced in the containers in accordance with MIL-STD-1186 using fiberboard conforming to PPP-F-320, Class Weather-Resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B. The tables shall be packed, and blocked and braced as specified in 5.2.1 for Level A, except the boxes shall be Domestic Type.

5.2.3 Commercial. The tables shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The drawing table is intended for use in surveying Equipment Set 6675-97-CL-E45.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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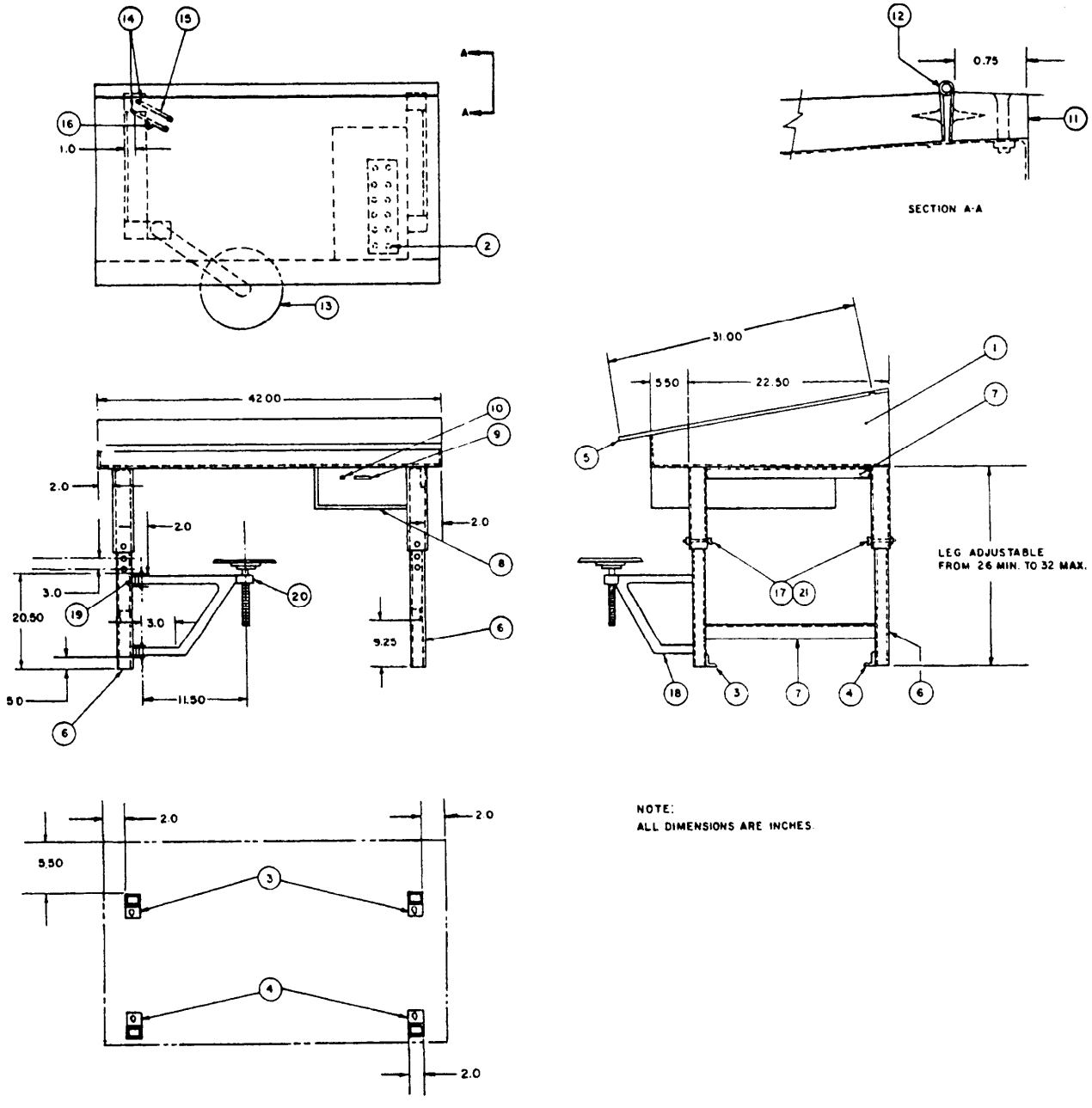
Preparing activity:
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Review activity:
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Project 6675-A226



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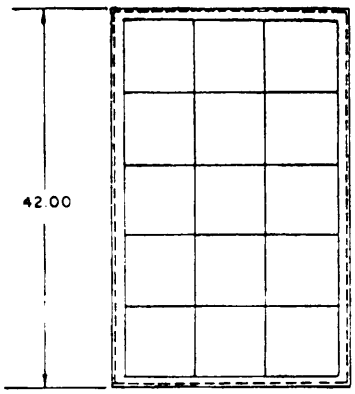

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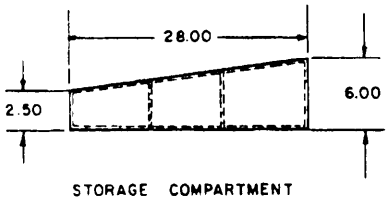
FIGURE 1. Table, drawing, metal base.

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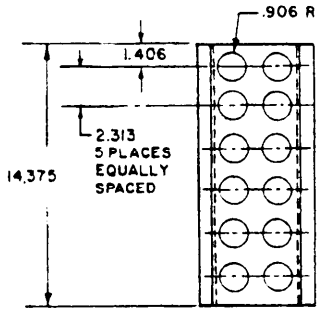


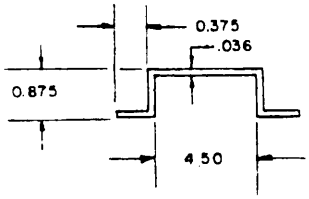
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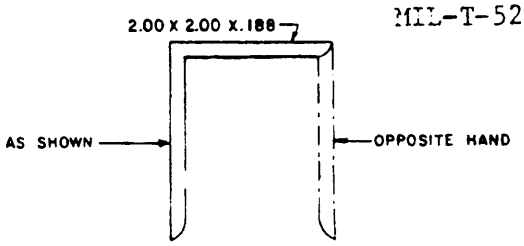



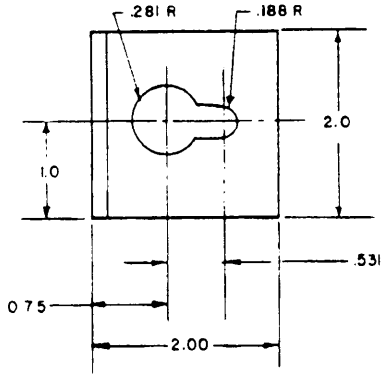

STORAGE COMPARTMENT




INK TRAY




KEY HOLE ANGLE

| FIND NO. | QUANTITY | DESCRIPTION |
|---|---|---|
| 1 | 1 | STORAGE COMPARTMENT |
| 2 | 1 | INK TRAY |
| 3 | 2 | KEY HOLE ANGLE |
| 4 | 2 | KEY HOLE ANGLE (OPPOSITE HAND) |
| 5 | 1 | DRAWING BOARD |
| 6 | 4 | LEG, ADJUSTABLE |
| 7 | 4 | BRACE, SIDE |
| 8 | 1 | DRAWER |
| 9 | 1 | DRAWER PULL |
| 10 | 1 | CATCH, PUSH BUTTON |
| 11 | 1 | STRIP |
| 12 | 1 | HINGE |
| 13 | 1 | SEAT, STOOL, SWIVEL |
| 14 | 2 | FOOTMAN LOOP |
| 15 | 1 | STRAP, WITH BUCKLE |
| 16 | 1 | STRAP, WITH METAL TIP |
| 17 | 4 | BOLT |
| 18 | 1 | BRACKET, STOOL |
| 19 | 2 | HINGE, STOOL |
| 20 | 1 | ROTARY MECHANISM, STOOL |
| 21 | 4 | WING NUT |

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FIGURE 2. Storage compartment, key hole angle, and ink tray.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-52820A(ME) Table, Drawing, Steel

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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TELEPHONE NO.

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